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RESEARCH ARTICLE

Compact and Affordable Raspberry Pi-Based Multimodal Imaging System With Remote Monitoring Capabilities for Bioimaging Applications

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ABSTRACT

The rapid advancements in smartphone camera features and accessibility have garnered significant attention for smartphone-based imaging systems in recent years. However, due to variations in dimensions and camera positions across different phones, they often suffer from compatibility issues. In the present work, these issues have been addressed by using a Raspberry Pi single-board computer (SBC) as the platform for the development of a multimodal microscopic system. The proposed imaging system can be remotely accessed and controlled via Wi-Fi from a computer, tablet, or smartphone, thus offering greater flexibility compared to smartphone-based imaging systems. The system offers four imaging modalities: bright-field (BF), dark-field (DF), oblique illumination (OI), and differential phase contrast (DPC), all on a single platform without requiring any additional optical components. Developed at an affordable cost of ~$122 using 3D printing and readily available components, the system achieves a noteworthy optical resolution of 1.64 μm, comparable to a laboratory microscope with a ×10 objective. Its imaging capabilities have been validated against a research-grade microscope, demonstrating its potential as a reliable and cost-effective tool for sectors in need of accessible microscopy solutions.

1 | Introduction

Optical microscopes are widely recognized as the benchmark for analyzing samples in diverse fields, that includes clinical laboratories, educational research laboratories, food industries, and pharmaceutical industries, and so forth. The use of microscopic systems to examine biological samples has driven major advancements in prognosis, research, and disease treatment, greatly contributing to the healthcare sector in terms of diagnosing a broad range of illnesses (Murphy and Davidson 2012; Malacara et al. 2003; Antony et al. 2013; Ozcan 2014). Among the various optical imaging techniques, BF imaging is the most widely used mode of imaging for viewing stained and colored microscopic samples, where the contrast in the microscopic image is generated by light absorption at different dense points of the specimen. However, the BF mode of imaging suffers from poor contrast quality when it comes to optically translucent or unstained colorless specimens. For such samples, different contrast-enhancing imaging techniques such as DF, OI, or DPC are often preferred (Mertz 2019; Boas et al. 2016; Zuo et al. 2016; Liu et al. 2014). The implementation of these contrast-enhancing techniques requires additional optical components and a

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Summary

- Low-cost, handheld Raspberry Pi-based microscopic system with BF, DF, OI, and DPC imaging modes.
- Supports remote viewing across devices for flexibility and accessibility.
- OLED panel serves as both an optical source and spatial light modulator.

spatially modulated illumination system, which increases the cost and complexity of the imaging platform; thus, limiting their feasibility in resource-limited areas.

Numerous groups over the past decade have reported various simplified microscopic imaging systems and novel contrast-enhancing techniques by reducing the requirement of additional optical and hardware components. This has been possible due to the availability of various low-cost consumer electronic devices such as smartphones, LEDs, Arduino microcontroller boards, and so forth. Using the readily available components, one can develop portable, cost-effective imaging and sensing devices, with smartphones being the most preferred platforms for such systems due to rapid advancements in their hardware and software capabilities (Rabha et al. 2019, 2021, 2022; Das et al. 2022; Banik et al. 2020, 2021; Kholafazad-Kordasht et al. 2021; Shu et al. 2021; Di Febo et al. 2021; Aryal et al. 2021; Christopher et al. 2025; Zeng et al. 2018; Ogasawara et al. 2019; Jung et al. 2017; Karki et al. 2024; Zhang et al. 2024; Lee and Yang 2014). Zeng et al. reported the development of a single-mode bright-field microscopic imaging device on a smartphone for cell counting using a custom-designed microfluidic chip. A ball lens was utilized as an objective lens for functioning as a microscope, and automatic or manual counting was possible using a custom-developed android application (Zeng et al. 2018). The majority of the reported works on the smartphone platform have demonstrated the functioning of a singular imaging modality, like BF or fluorescence mode. However, several research papers have shown that smartphone-based systems can be prototyped for various microscopic modalities. Ogasawara et al. demonstrated multimodal smartphone microscopy using an array of 37 LED bulbs for BF, DF, DPC, and Rheinberg illuminations. Rheinberg illumination enabled simultaneous acquisition of both BF and DF images using two color light sources (Ogasawara et al. 2019). Similarly, Jung's group reported a smartphone-based multicontrast microscopic system capable of imaging in three different modalities, including DPC by making use of an LED matrix illumination (Jung et al. 2017). Again, Zuo's research group has designed a multimodal microscope with a liquid crystal display (LCD) panel in place of a condenser diaphragm on a laboratory microscope. The LCD has been used as the spatial light modulator (SLM) and several imaging modalities have been realized by generating distinct illumination patterns on the LCD (Zuo et al. 2016). While the smartphone-based optical analytical systems offer several key benefits such as cost-efficiency, user-friendliness, and portability, the diverse physical dimensions and rear camera positions across different phone models restrict the universal adaptability of the optical setup on different smartphones. An imaging system developed for one specific smartphone may not be suitable for others; thus, restricting the use of phone-based analytical systems only to a specific phone model. This lack of flexibility reduces the system's adaptability across different smartphones, hindering its accessibility for people carrying different smartphone models. To address this issue, several groups have reported the development of a universal holder compatible with different shapes and sizes of smartphones. Bogucki et al. reported a three-dimensional (3D)-printed smartphone adaptor meant for dual-beam spectrophotometric studies. The designed universal mounting system allowed a wide variety of smartphones to be coupled to a spectrophotometer (Bogucki et al. 2019). Vidal et al. reported the working of a 3D-printed smartphone-based spectrophotometer with a universal smartphone holder that could be interfaced with different kinds of smartphones, irrespective of their size and shape and the position of the camera (Vidal et al. 2021). These universal holders, despite offering the advantage of compatibility features, suffer from relatively complex design due to the inclusion of numerous moving parts, which subsequently could reduce their durability and reliability over extended periods of use.

Few works on the Raspberry Pi platform as a microscopic system have been reported (Watanabe et al. 2020; Deglint et al. 2016; Diederich et al. 2020), but no work on the design of a handheld, portable multimodal imaging platform using this platform has been reported till date. Herein, we demonstrate for the first time a multimodal microscopic imaging system using the Raspberry Pi SBC platform, which has the ability to capture images in four different modes, namely the BF, DF, OI, and DPC modes of imaging. Readily available off-the-shelf consumer-grade optoelectronic parts have been used to design the proposed imaging platform. This approach obviates the key drawbacks associated with smartphone-based imaging systems, thereby improving the flexibility and universality of the designed microscopic system. The proposed Raspberry Pi-based imaging device can be accessed on any display device remotely using Wi-Fi on account of the wireless capabilities of the SBC, making it a standalone system that is not limited to a specific device like a smartphone or computer. The adaptability and remote viewing capabilities of the imaging device have been demonstrated by operating the system on three different devices: a smartphone, a tablet, and a computer. Compared to a typical smartphone-based microscopic system, the proposed Raspberry Pi SBC imaging platform is cost-effective, making it a viable and promising alternative for implementation in resource-constrained regions. The SBC system is highly versatile and adaptable, allowing users to connect the Raspberry Pi board to a wide range of electronic circuits, sensors, and various types of camera modules. For the designed system, a mini-OLED panel serves both as an optical source and an SLM, producing precise spatial patterns for the multimodal imaging platform. The system's performance and applicability have been validated through imaging a range of different samples such as standard microbeads, Leishman-stained blood smear, hematoxylin and eosin (H&E)-stained nerve cells, and three different biological samples, and the results were compared with the standard laboratory microscope. The developed imaging system is relatively inexpensive, compact, and portable as it is entirely housed in a 3D-printed enclosure.

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2 | Materials and Methods

2.1 | Microscopic System Design

Figure 1 illustrates the workflow implemented in this study, highlighting the key steps from the schematic representation of the system to the operation of the proposed microscopic setup, including image acquisition and remote accessibility. The optical layout employed in the designing of the imaging system is shown in Figure 2a. The designed system is based on a 3f optical configuration, where a compact and simple optical design implementation is possible. The built-in lens of the Raspberry Pi camera module (v3 standard camera module) acts as a tube lens. A rear camera lens of a smartphone (Redmi K20 Pro) has been used as an objective lens for the proposed microscopic system. It is attached to the Raspberry Pi camera module to form a finite-conjugated optical system. This optical design approach enables the fabrication of a simple and economical microscopic system on the Raspberry Pi platform. The proposed microscopic system is a standalone unit where the key components of the imaging system consist of an OLED display panel serving both as an optical source and an SLM, imaging optics, Raspberry Pi SBC, a translational slide holder, a focusing knob, and 3D-printed holders and enclosure. The proposed platform further ensures affordability and portability which can be developed in a common laboratory environment. The dimension of the designed system is 73 mm × 108 mm × 88 mm, and the total weight is measured to be ~225 g including the Raspberry Pi SBC. An X-Y translational 3D-printed slide holder is designed to load a standard microscopic slide into the set-up, and a 3D-printed focusing knob holder is attached to the setup for movement of the slide holder along the Z-direction for focusing of the sample. Figure 2b shows the photograph of the prototype of the fabricated system. The 3D design of the different parts of the proposed imaging system, including the sample stage and the illumination block, is shown in Figure 2c. Figure 2d shows the different illumination patterns that have been generated on the OLED panel to obtain different imaging modalities for the present microscopic system.

The Raspberry Pi 5 SBC platform consists of a powerful 2.4 GHz 64-bit CPU with 8 GB RAM and built-in Wi-Fi and Bluetooth capabilities (Raspberry Pi Ltd 2024a). This SBC is extremely versatile and flexible as it consists of several GPIO pins and connectors that enables user to connect the Pi board to different electronic circuits and sensors. It also supports the integration of different kinds of camera sensors onto the Pi board. We have utilized the Raspberry Pi camera module 3 (standard v3 module) as the main imaging sensor

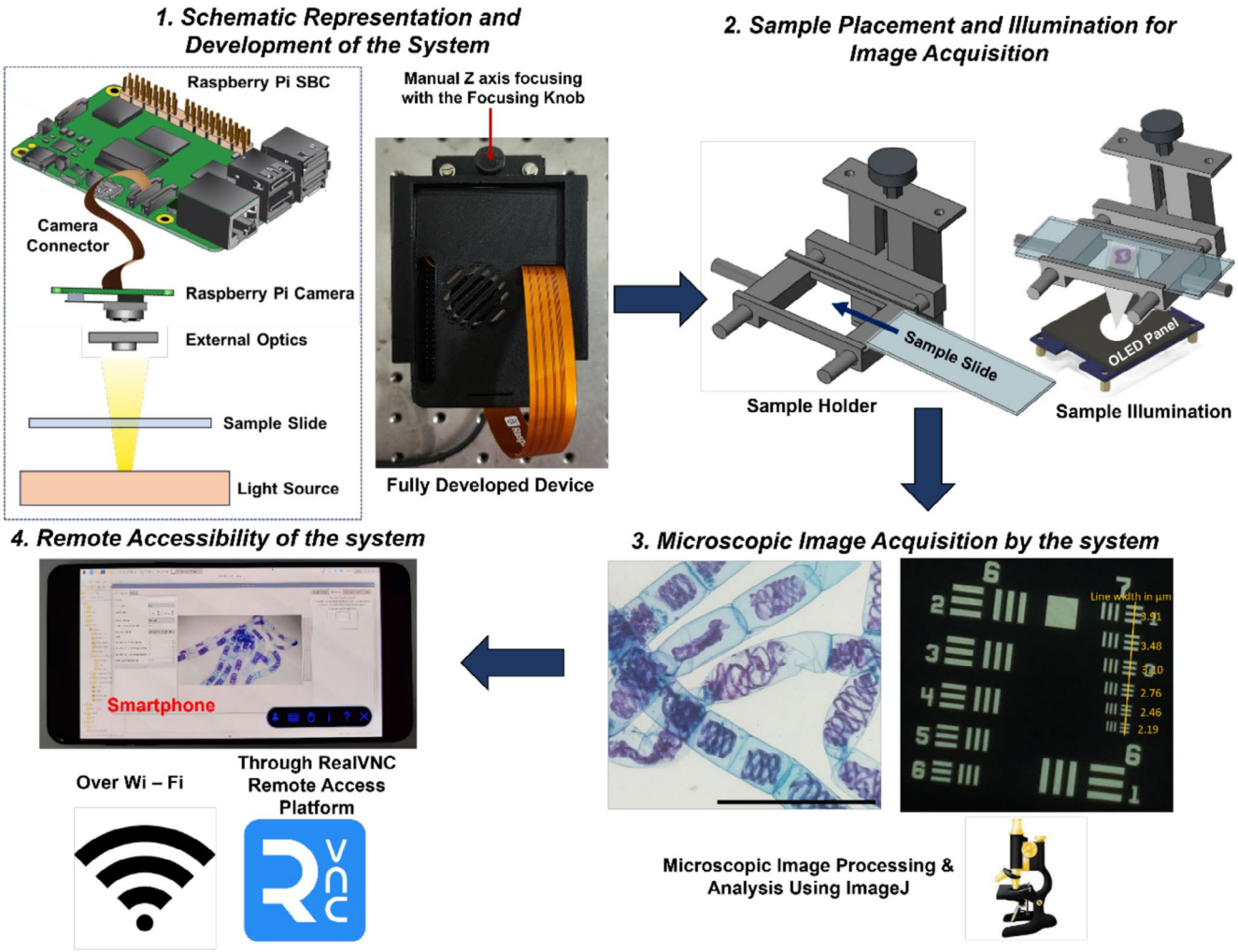

FIGURE 1 | Workflow illustrating the key stages involved in the development and operation of the proposed microscopic system.

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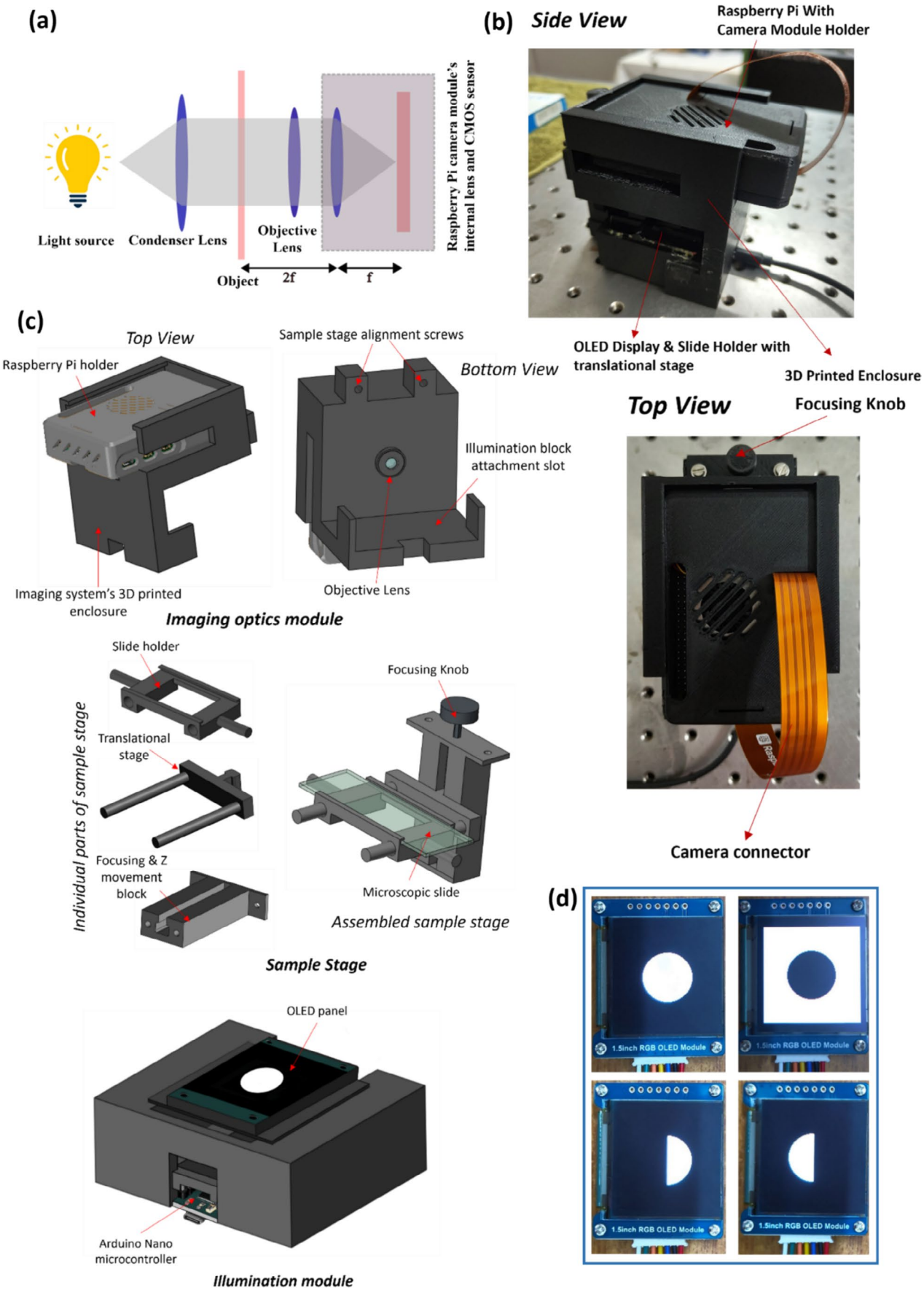

FIGURE 2 | Legend on next page.

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FIGURE 2 | Design of the Raspberry Pi-based imaging system. (a) The ray diagram for the 3f optical configuration. (b) The photograph of the developed microscopic imaging system from the side and top view angle. (c) The 3D design of the components integrated in the imaging system, which constitutes the imaging optics block followed by the sample stage and the illumination module. (d) The different illumination patterns on the OLED panel generated for obtaining the multiple imaging modes.

TABLE 1 | List of components used for the development of the proposed imaging system.

Component	Source	Purpose	Cost (in $)
Raspberry Pi 5	Online electronics store (Robu.in)	SBC module	67.0
Pi Camera standard v3	Online electronics store (Robu.in)	Imaging sensor	34.0
Smartphone external camera lens	Local Supplier	Objective Lens	7.0
RGB OLED display	Local Supplier	Light Source and SLM	10.0
Arduino Nano board	Online electronics store (Robu.in)	OLED illumination controller	2.0
3D printed holders	Laboratory fabricated	Enclosing the components of the system	1.00
Miscellaneous	Local market	Focusing and slide movement	1.00
		Total cost	122.00

connected to the Raspberry Pi 5 board. The v3 module has 12 MP (Megapixel) Sony IMX708 COMPLEMENTARY METAL OXIDE SEMICONDUCTOR (CMOS) imaging sensor with sensor dimension of 7.4 mm diagonal and integrated with an optical lens of focal length (FL) 4.74 mm (Raspberry Pi Ltd 2024b). It has an aperture size of f/1.8 and numerical aperture (NA) 0.21. The pixel size of the imaging sensor is 1.4 μm × 1.4 μm. Hence, to resolve an object of size 1.5 μm, the optical magnification required would be at least $M \geq 1.87$ ($M \geq \frac{2 \times \text{pixel pitch of the sensor}}{\text{Required resolution}}$) (Rabha et al. 2022). The designed optical set-up provides the essential optical magnification of 1.87 needed to resolve 1.5 μm objects by forming object to image size ratio of 1:1.87.

To develop the microscopic system in 3f optical configuration, an objective lens of small FL is required. Different micro-objective lenses such as ball lenses, aspherical lenses, and achromatic lenses are commercially available for such purposes. However, in the present microscopic system, the wide-angle rear camera lens of a smartphone-Redmi K20 pro has been used as an objective lens. The primary motive behind using this lens is that it is made up of multiple refractive index elements that are well corrected to reduce possible aberrations and distortions in the final image that may otherwise be present in the aforementioned micro-objective lenses. The Redmi K20 Pro wide-angle lens has a FL of 2.04 mm with an aperture of f/2.4 and NA 0.21. This micro-objective lens is placed in reverse orientation and is aligned with the optic axis of the camera module lens of the Raspberry Pi SBC to form a finite-conjugated optical imaging system. All components used in the setup are low-cost, readily available, consumer-grade items. The breakup cost of all the components involved in the development of the prototype of the present imaging system is given in Table 1. To the best of our knowledge, no commercial multimodal microscopic system is available at this price, thus signifying the impact of the present work along with the flexibility features as narrated above.

2.2 | OLED Illumination System and Image Acquisition

The optical source of the present imaging system is a low cost RGB OLED panel (IC = SSD1351) of dimension 1.5 in. and pixel array of 128 × 128. The display panel is placed at a distance of 20 mm from the specimen plane to illuminate the sample. The OLED panel is fully programmable and can generate different colors and binary patterns, suitable to use it both as an optical source and as SLM for the proposed multimodal imaging system. The emission spectral characteristic of the illumination panel while displaying a bright circle is recorded by a CCD spectrometer (Model—CCS200). Figure 3a shows the obtained spectral response of the OLED panel. We use an Arduino Nano microcontroller to program and control the display patterns of the OLED panel. The display panel can be powered from a powerbank or from any device such as a smartphone or computer through Raspberry Pi module as an interface. In the present study, we have used smartphone's internal battery to power the OLED panel using a Universal Serial Bus On-The-Go (USB-OTG) cable. An open-access android application "Serial USB Terminal" is used to control the display patterns of the OLED panel. The different illumination patterns generated on the OLED panel are shown in Figure 2d. For BF mode, a bright white circle is generated on the illumination panel, while turning off all other pixels outside the generated circle making the illumination as a bright source against a dark background. The diameter of the bright circular spot can be tuned according to the NA of the objective and this adjustment is analogous with the alteration of

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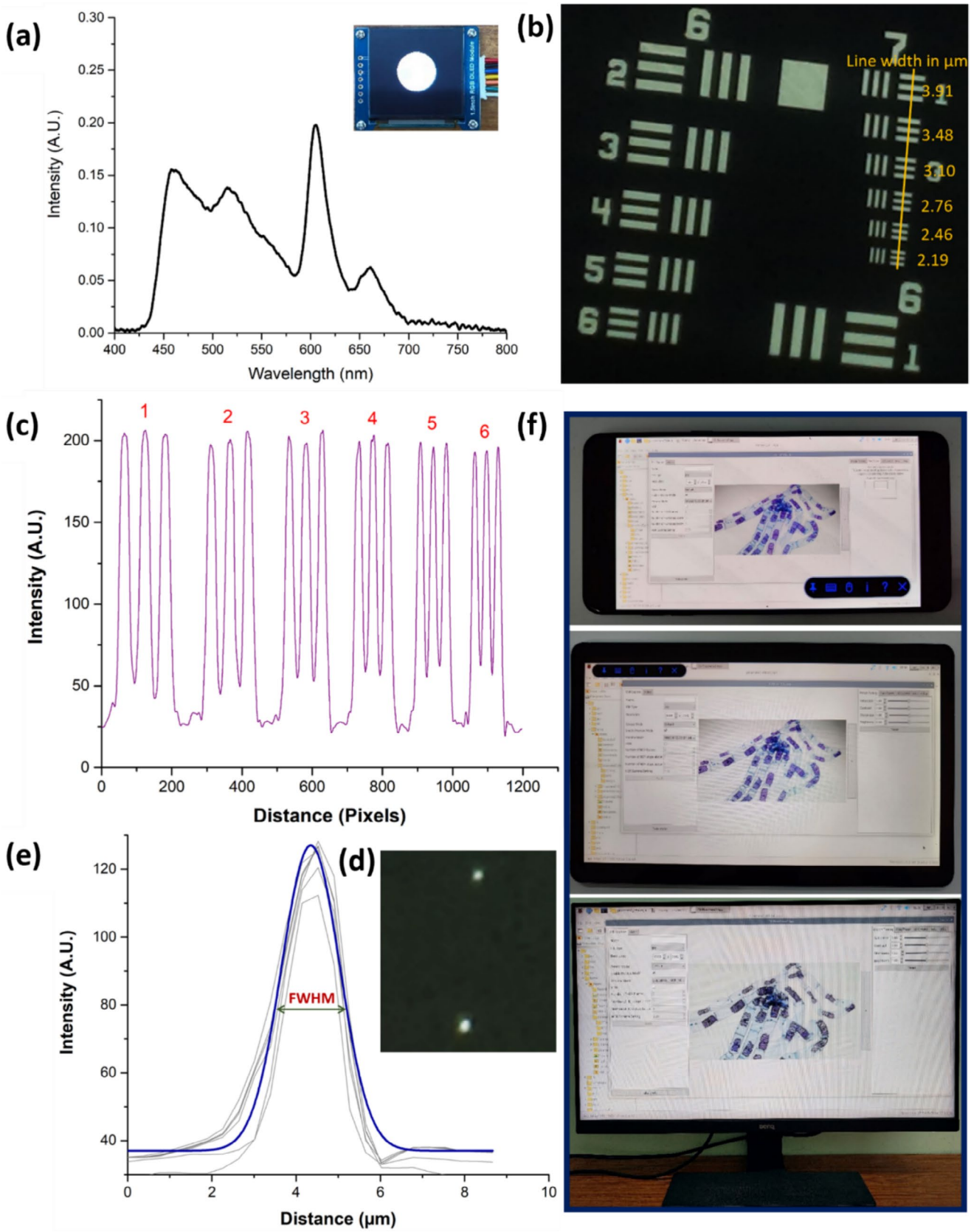

FIGURE 3 | Optical characterization and image acquisition of the Raspberry Pi-based imaging system. (a) The spectral graph of a bright circle displayed on the OLED panel along with the pattern. (b) The captured image of a standard 1951 USAF resolution test target using the designed system, with Group 7 element 6 bars clearly visible, and (c) the intensity profile of Group 7. (d) The image of 1 µm standard beads captured by the system, and (e) the FWHM measurement of the point spread function (PSF) of the system through the imaging of several 1 µm beads and taking the average values of the microbeads. (f) The remote wireless accessibility of the Raspberry Pi imaging system on three different devices—Samsung C9 Pro smartphone, Samsung S6 lite tablet, and a BenQ monitor, which shows the standalone universal feature of the imaging system.

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condenser aperture in traditional optical microscope. For DF mode of imaging, a dark circular pattern is generated on the OLED display while the surrounding background pixels are turned white as seen in Figure 2d. The diameter of the dark circle has been fixed at a value just greater than the objective lens' NA so that it would restrict the central beam to pass through an objective lens, which resembles physically adjusting the aperture size of the DF stop present in the condenser in a conventional microscope. Another contrast-enhancing imaging technique is the OI mode, where a pseudo 3D image of the specimen providing a higher contrast of the specimen can be viewed. As can be seen in Figure 2d, this mode of imaging can be achieved by generating either a right or left white semicircle on a dark background on the display panel. This is similar to using an opaque light stop having a small slit placed beneath the condenser lens in a conventional microscope. Another significant contrast-enhancing imaging technique is the phase contrast mode of imaging. Two semicircular patterns complementary to each other are generated on the illumination panel and the corresponding images are captured by the system. The captured images along the two different semicircular orientation contain the phase information of the sample along those axes and this information can be retrieved by calculating the normalized value of the difference in the intensities of the two images by using the following equation:

$$I_{\mathrm{DPC}} = \frac{I_1 - I_2}{I_1 + I_2} \tag{1}$$

where, I_1 and I_2 are the intensities of the right and left semicircular illuminated images. A python script has been implemented to compute the normalized intensity values and to generate the final DPC image. The functioning of the present imaging system relies greatly on the precise alignment of the optical design of the system. It is imperative that the pattern displayed on the illumination panel is properly positioned with the optical axis of the proposed imaging setup.

Again, the camera module of the Raspberry Pi SBC can be accessed using an open-source camera application. The application has been developed on the picamera2 python library. Using this application, the designed setup is capable of capturing the images of the specimen under examination (Raspberry Pi Ltd 2024c). This camera application provides a plethora of advanced camera control features, like exposure time, ISO control, white balance, brightness control, and so forth. It can also capture images in RAW DNG (digital negative) format. To capture the images in different imaging modalities, the ISO and the exposure time are to be adjusted accordingly. To capture images in different modes in imaging, the ISO has been maintained in the range 100–400 and the exposure time has been set in the range between 200 ms to 2 s. On account of the wireless capabilities of the Raspberry Pi SBC, the captured images can be accessed remotely on different devices such as a smartphone, a tablet, or from a computer. This offers a unique advantage of the present imaging device as being flexible and universal and hence, not restricted to a single device such as a

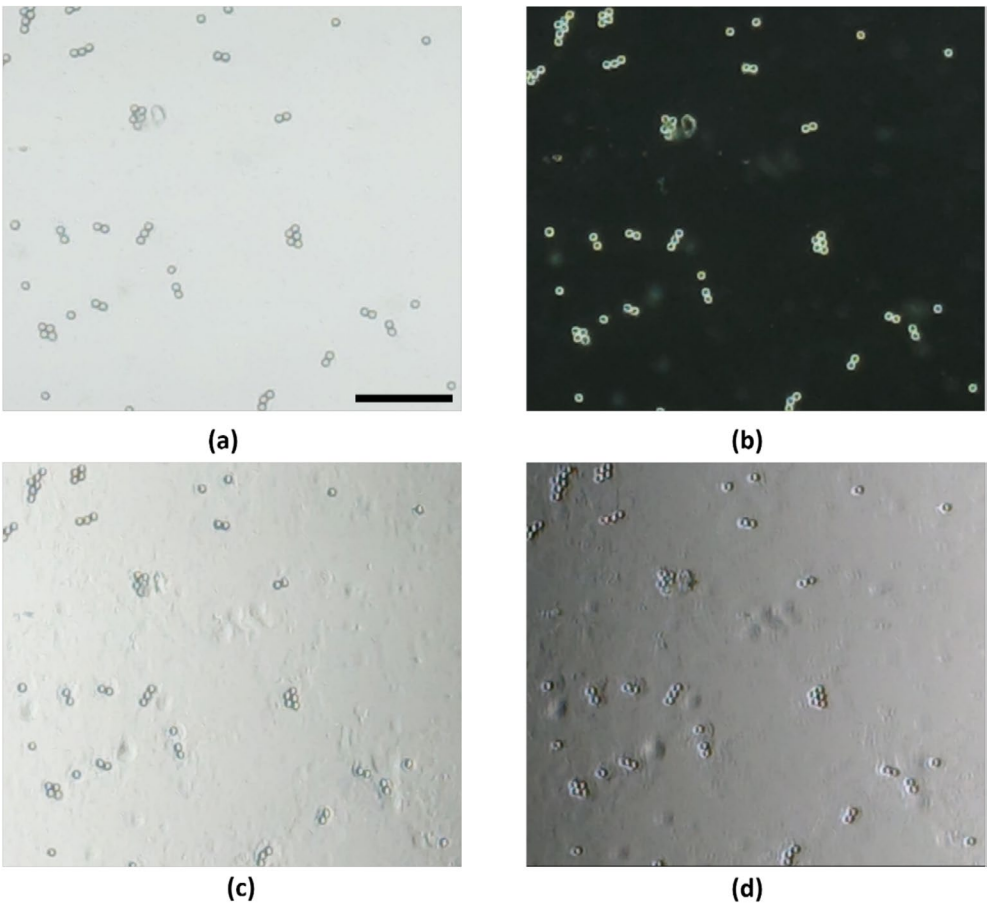

FIGURE 4 | Images of standard 5 μm silica beads captured by the Raspberry Pi microscopic system in different imaging modalities: (a) BF, (b) DF, (c) OI, and (d) DPC modes (scale bar is 50 μm).

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smartphone as reported elsewhere. The remote accessibility of the imaging system on different devices is shown in Figure 3f. Although remote accessibility provides flexibility to operate the microscopic device, it does come with certain limitations such as latency and frame rate reduction. To quantify latency, we conducted a stopwatch-based assessment comparing the time delay between a local frame and its corresponding appearance on a remotely accessed device. This analysis revealed that an average delay of 332 ms occurred between these two situations. The anticipated delay is primarily attributed to cumulative delays introduced due to camera sensor readout, image processing, encoding, Wi-Fi transmission, and rendering on a remote display device. While this latency may be detrimental for monitoring live cells and their interaction, it remains within an acceptable range for static microscopy imaging, where real-time responsiveness is less critical. To evaluate frame rate degradation during remote access, we compared the camera's native output with the actual frame rate observed on the remotely connected device. The Raspberry Pi camera was operated at a frame rate of 30 frames per second (FPS). However, through a frame-by-frame analysis of a remote session recording, we observed an effective remote viewing frame rate of ~19 FPS. This decrease is primarily attributed to encoding overhead, network latency, and rendering limitations. Despite this reduction, the received frame rate is sufficient for sample monitoring and imaging. Minor compression artifacts were occasionally observed during remote viewing, particularly during rapid sample movement or alignment. However, these did not result in any significant loss of structural detail or affect the visibility of key features. To process the captured images and to calculate various parameters such as field of view (FoV), intensity profile, and incorporation of scale bar to the images, an open-source computer software ImageJ has been implemented.

3 | Results and Discussion

3.1 | Optical Characterization of the System

Prior to the capture of images from the specimens, the microscopic parameters of the designed imaging system have been evaluated. Owing to the use of polylactic acid (PLA) as 3D printing filament material, the designed optical system is fairly robust. The finite-conjugated optical system provides an optical magnification of ×2.37, which is the ratio of the FL of the Raspberry Pi v3 camera module lens (FL = 4.74) to the FL of the external micro-objective lens (FL = 2 mm). Again, depending on the device where the imaging system would be viewed, the displayed image would also have a digital magnification factor ($M_d = \frac{\text{Screen size}}{\text{Sensor Size}}$). In the present study, a smartphone, a tablet and a monitor have been used to access the system and display the microscopic image whose display sizes are measured to be 6, 10.4, and 23 in. respectively. The Raspberry Pi v3 camera module has an image sensor of 1/2.43 in. (7.4 mm sensor diagonal), so the digital magnification for these three devices would be M_d = ×20.6, ×35.7, and ×78.9, respectively.

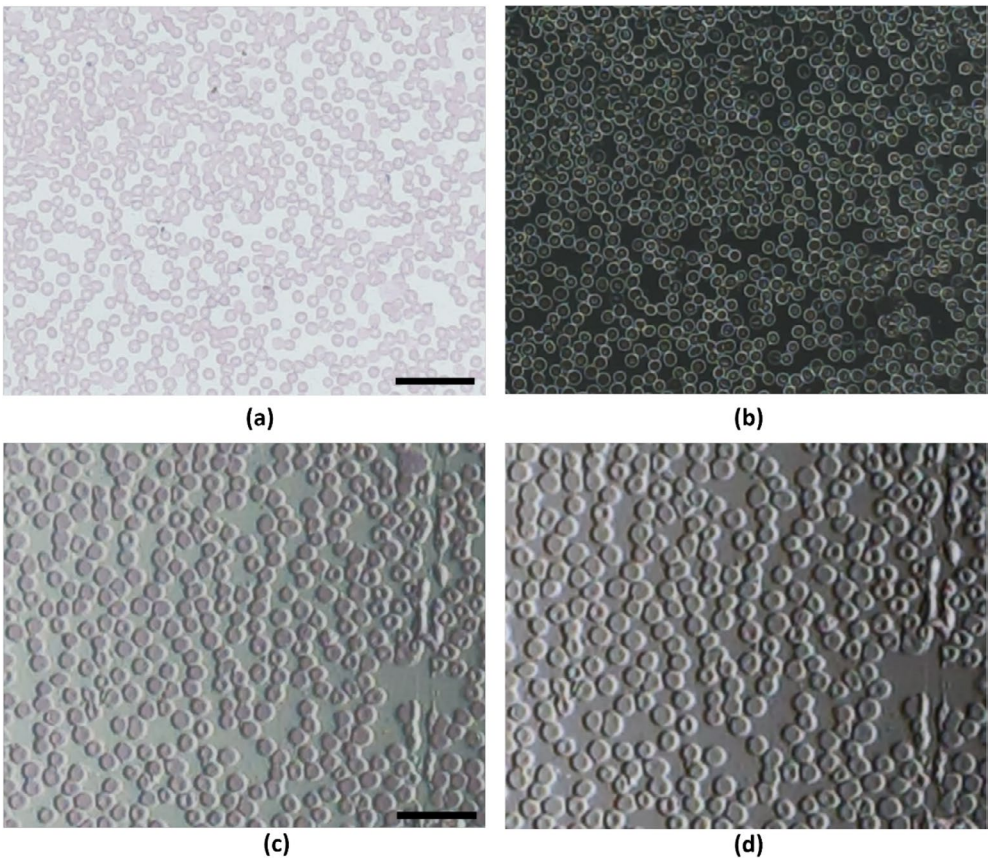

FIGURE 5 | Performance evaluation of the designed microscopic system through imaging of Leishman-stained thin blood smear slide in (a) BF, (b) DF, (c) OI, and (d) DPC imaging modes. Scale bar: 50 μm in (a, b) and 30 μm in (c, d).

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The NA of the external objective lens is 0.21, which theoretically gives a diffraction resolution limit of 1.59 μm ($Rth = \frac{0.61\lambda}{NA}$, $\lambda = 550$ nm). To obtain an experimental lateral resolution of the present imaging system, a 1951 USAF test target has been imaged in BF mode. This test target consists of several groups with different elements consisting of horizontal and vertical bars indicating different resolution. The image of the test target captured by our system is given in Figure 3b. It has been observed that the developed imaging system can spatially distinguish the smallest line element, that is Group 7 Element 6 of the test target, with ease. Figure 3c shows the plot profile of the intensities of Group 7 Element 6 bars, indicating excellent contrasting ability of the proposed microscopic system which corresponds to a density of 228.1 line pairs per millimeter (lp/mm), equivalent to a resolution of 2.19 μm. Again, to further evaluate and obtain the lateral optical resolution of the proposed setup, we estimate the point spread function (PSF) of the imaging system. The imaging of 1 μm beads in DF mode has been captured with our designed system. Figure 3d shows the image of the 1 μm beads recorded by the imaging tool. As shown in Figure 3e, the Gaussian fitted intensity profile graph provides the full-width half maximum (FWHM) of the PSFs of different 1 μm beads in lateral direction captured by the designed imaging system. The average intensity distribution of the imaged beads yields an experimental FWHM value of 1.64 μm, indicating that the achieved optical resolution of the present microscopic system is

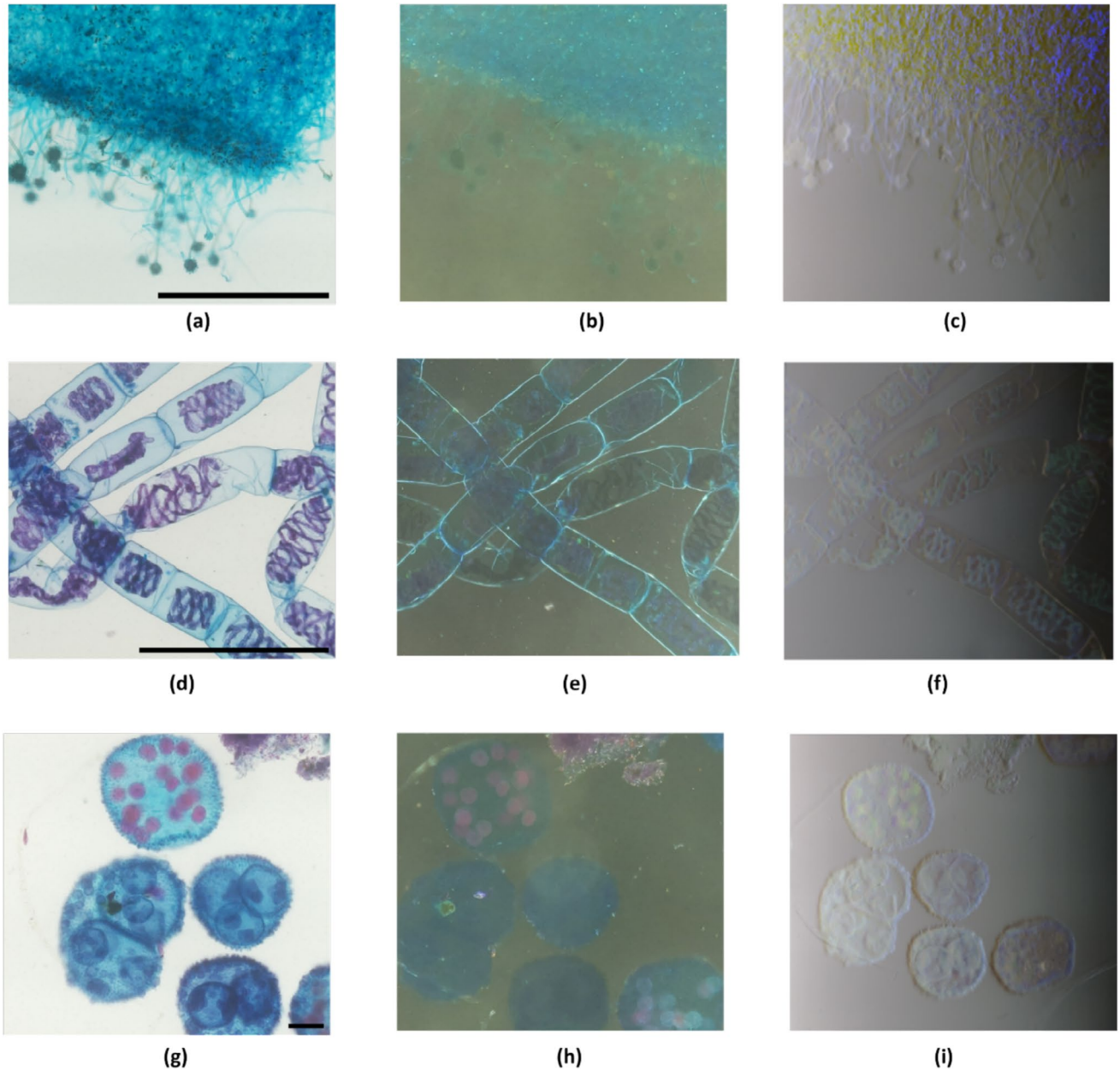

FIGURE 6 | Application of the Raspberry Pi microscopic system in imaging biological samples under different imaging modes. (a–c) The images of the *Penicillium* fungi captured in BF, DF, and DPC modes, respectively. (d–f) showcases images of Algae *Spirogyra* in BF, DF, and DPC modes, respectively, and (g–i) displays Algae *Volvox* images captured in the three modes, BF, DF, and DPC, respectively. Scale bar: 50 μm in (a–f) and 200 μm in (h, i).

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approximately 1.64 μm. To evaluate the imaging performance, we calculated the Nyquist-limited sampling resolution. Initially, considering only the Raspberry Pi Camera with a pixel pitch of 1.4 μm and no additional optical magnification, the Nyquist criterion would imply that the minimum resolvable feature size is twice the pixel pitch, that is, $2 \times 1.4\,\mu m = 2.8\,\mu m$, which is clearly insufficient for the claimed resolution. However, the designed imaging system has an effective optical magnification of ×2.37. This magnification reduces the effective sampling interval in object space to ~0.59 μm per pixel (Effective pixel size $= \frac{\text{Pixel size}_{\text{sensor}}}{\text{Magnification}} = \frac{1.4\ \mu m}{2.37} = 0.59\ \mu m$). According to the Nyquist criterion, to resolve a feature of 1.64 μm, the pixel spacing must be ≤0.82 μm in object space. Since 0.59 μm is well below this threshold, the system comfortably satisfies the Nyquist condition and is thus capable of resolving 1.64 μm features without aliasing artifacts. Additionally, the external lens provides a theoretical diffraction-limited resolution of 1.59 μm. This optical resolution aligns well with the claimed experimental system performance. In the next step another two important parameters contrast and the signal-to-noise ratio (SNR) of the microscopic system have been evaluated. The contrast can be estimated as

$$C = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} \tag{2}$$

where, I_{max} is the maximum intensity of the specimen and I_{min} is the minimum intensity of the background, and the SNR can be calculated as

$$\text{SNR} = \frac{I_{specimen} - I_{bg}}{\sigma_{bg}} \tag{3}$$

where, $I_{specimen}$ is the average intensity value of the specimen, I_{bg} is the average intensity value of the background and σ_{bg} is the standard deviation value of the background. With 5 μm bead images, the BF mode of imaging provides a contrast of $C = 0.068$ and SNR = 6.244. These values infers that the BF mode does not provide sufficient contrast to the images which result in lack of details in the images. For the DF and DPC modes, the contrast and SNR values are {C, SNR = (0.655, 44.48) and (0.17, 6.42)} respectively. Compared to the BF mode, the contrast and SNR values are observed to be enhanced significantly in the DF and DPC modes for the proposed microscopic system. Depth of field (DoF) is another important parameter of a microscopic imaging system. In microscopy, DoF refers to the range of distance within the specimen that appears acceptably sharp in an image; that is basically, it is the depth of the specimen that remains in focus at a particular time. It is related to the NA of the objective lens in the imaging system, with higher NA providing a considerably smaller DoF. For the designed imaging system, the DoF is

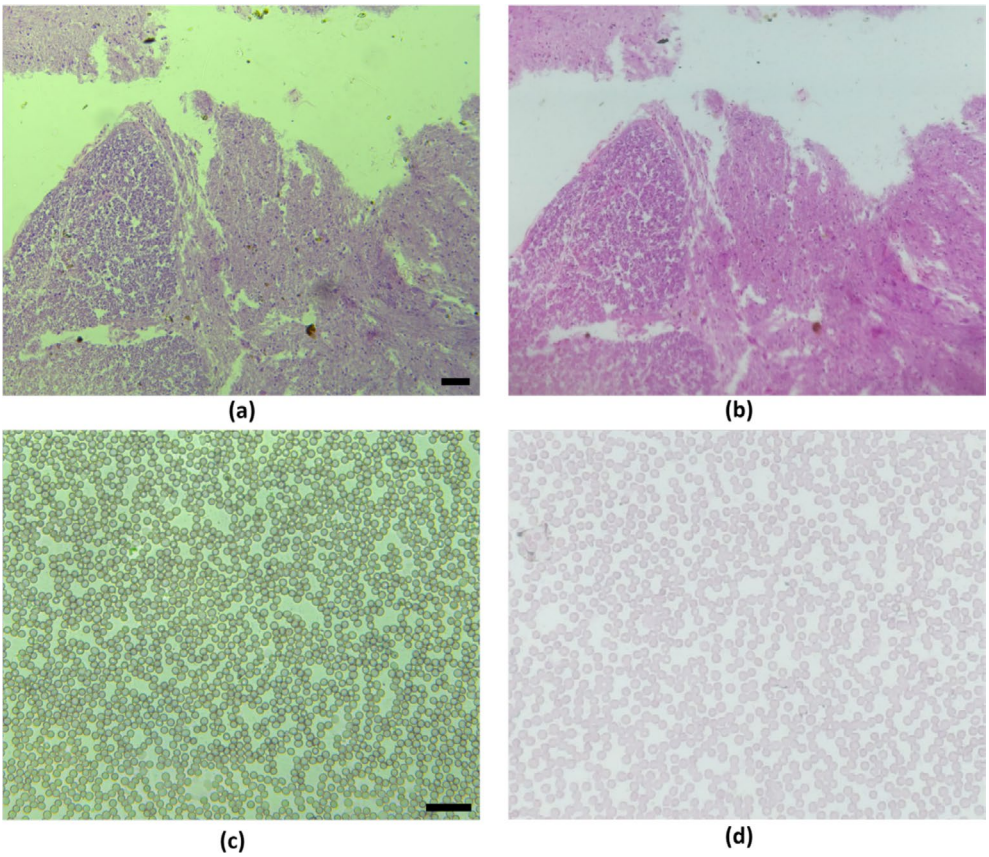

FIGURE 7 | Image acquisition and comparison of Raspberry Pi microscope with laboratory optical microscope (×10, NA = 0.25). (a) The image of H&E-stained nerve cell section taken by laboratory microscope and (b) the image captured by Raspberry Pi microscope. (c, d) The images of Leishman-stained blood smear slide captured by laboratory microscope and Raspberry Pi microscope. Scale bar is 50 μm.

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TABLE 2 | Comparison of multimodal microscopic systems across different parameters.

Imaging platform	Optical magnification	Resolution	Contrast and SNR	Compatibility	Development cost	References	Remarks
Smartphone	×1.66	1.7 μm	Not disclosed	Restricted to operation on a single smartphone	$ 290	Rabha et al. (2022)	Total cost is calculated excluding the smartphone utilized. Consequently, the actual cost is significantly higher. The system's operation is restricted to a specific smartphone model.
Raspberry Pi	Not disclosed (×10 objective used)	< 4.4 μm	Not disclosed	Operated via a desktop	Not disclosed	Watanabe et al. (2020)	Being developed on a laboratory bench, the system is relatively bulky and not suitable for in field applications.
Raspberry Pi/ Smartphone	Not disclosed (×4/×10/×20/×100 objectives used for different imaging modes)	0.6 μm/1.13 μm (with ×100), 2.2 μm (with ×10)	Not disclosed	Operated via a Smartphone and a Raspberry Pi display	€ 100–600 depending on the imaging mode	Diederich et al. (2020)	Although the reported platform offers an open-source 3D-printed modular design, it involves certain complex components, and the operational costs for implementing different imaging modes are relatively high compared to our system.

(Continues)

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TABLE 2 | (Continued)

Imaging platform	Optical magnification	Resolution	Contrast and SNR	Compatibility	Development cost	References	Remarks
Raspberry Pi	×2.37	1.64 μm	BF: {C, SNR = (0.068, 6.244)}; DF: {C, SNR = (0.655, 44.48)}; DPC: {(0.17, 6.42)}	Accessible and operable on multiple devices such as smartphones, tablets, laptops, or desktops	$ 122	Present work	System features universal compatibility, enabling operation across multiple devices. Extremely cost-efficient and portable.

calculated to be ±6.1 μm, which means that the sample is in perfect focus at a distance of 6.1 μm above and below the specimen plane and anywhere outside this boundary, the image begins to blur. The present imaging system has a FoV of 3122 μm diagonally, with the horizontal and vertical FoV measured to be 2721 and 1531 μm, respectively.

3.2 | Evaluation of the Imaging System With Biological Specimens

The designed multimodal imaging system's applicability has been demonstrated through the imaging of different biological specimens. Before that, imaging of standard 5 μm silica beads under different imaging modes has been evaluated. Figure 4 indicates the images of the microbeads captured in BF, DF, OI, and DPC modes of imaging with the present microscopic system. From these figures, it is clear that for all the considered modes of imaging, the system has been able to generate detailed images with sufficient contrast, where the shape and structure of the microbeads are clearly visible. Also, the debris present in the microscopic slide is clearly visible in DF, OI, and DPC modes. For OI and DPC modes, the images of the microbeads produce a pseudo 3D image with a shadow effect and provide a better structural similarity and enhanced contrast to the images compared to BF mode of imaging. Leishman-stained human blood smear slide has been acquired from Tezpur University Health Centre for image acquisition with the developed system. Figure 5 shows the images of the blood smear taken by the proposed setup. As can be seen, the BF mode provides a clear image of the blood smear with the RBCs visible in the image.

However, compared to BF modality, images that are captured in DF and DPC modes yield an improvement in the contrast produced by the specimen. Also, the OI mode produced relatively better contrast in the image compared to BF mode. One of the important unique features of the present microscopic system is that the different imaging modes can be switched seamlessly from one imaging mode to another by merely changing the display patterns displayed on the OLED panel. *Penicillium rubens* is a fungus widely known for producing the antibiotic penicillin. Figure 6a shows the image of the *Penicillium* fungi in BF mode, and Figure 6b,c represents the DF and DPC image of the same sample captured by the imaging system. The image clearly shows the visibility of mycelium, the long filamentous hyphae and conidiophores, the spore-bearing structures of the fungus specimen. All the imaging modes provide a good order of contrast to the *Penicillium* fungus specimen's images captured by the proposed system. Another microbial species that has been considered in the present study is the algae species *Spirogyra*, which is a free floating, commonly found algae in stagnant water bodies having a helical chloroplast structure in its body. The BF, DF, and DPC modes of images of the considered algae captured by our imaging system are shown in Figure 6d–f. The cell wall of the algae, along with the spiral-shaped chloroplast structures with small and dense pyrenoid structures within the chloroplasts inside the cells, can be clearly seen with the proposed microscopic system. Another microbial species-*Volvox* has been considered as a test sample for evaluation. *Volvox* is a species of freshwater green algae commonly known as globe algae because of its

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spherical shape. The imaging system captured the images of the said algal species in all the abovementioned modalities, which showed the structural parts of the *Volvox*, like the cell walls of the algae along with the spherical colonial cells with the daughter colonies present inside the *Volvox* colony. Figure 6g–i shows the images of algae *Volvox* captured by the developed imaging system. The considered stained specimens have been obtained from Biological Hub, India. Due to the procured sample slides being stained in nature, the DF and DPC images have less contrast features as compared to unstained samples, which provide much better contrast in these two modes. Owing to its compact size, the proposed imaging system has the potential for utilization as a diagnostic imaging tool for infield applications, which can be seen through the successful imaging of the fungi and algal species.

Finally, the performance of the proposed imaging system has been compared with the images captured by a standard laboratory microscope. Histopathological samples like H&E-stained nerve cell tissue and Leishman-stained blood smear samples have been considered for the performance evaluation comparison of the proposed imaging system. Figure 7a shows the image of the nerve tissue section taken by the optical microscope, while Figure 7b shows the same section image taken by the Raspberry Pi microscopic system. The figure clearly indicates that the performance of the developed Raspberry Pi based imaging system is comparable to that of a laboratory optical microscope. The blood smear images taken by the laboratory microscope and the proposed microscopic system are shown in Figure 7c,d. These captured microscopic images clearly show that the performance of the proposed imaging system is comparable to that of a traditional optical microscope. A similar work on the development of a multimodal microscopic system utilizing smartphone was reported by Rabha et al. (2022). However, the present system distinguishes itself by offering universal compatibility across various devices, greater cost-effectiveness, and better optical performance. A comparison of the performance of the present imaging system with that of Rabha et al. (2022) and similar Raspberry Pi-based imaging systems by Watanabe et al. (2020) and Diederich et al. (2020) is provided in Table 2.

The results obtained from the developed imaging system demonstrate that the Raspberry Pi SBC platform is a reliable alternative for developing miniature imaging tools. The proposed system is also highly versatile, capable of supporting additional imaging modalities such as fluorescence and polarization imaging by integrating optical components like filters and polarizers, which will be explored in future studies.

4 | Conclusion

A standalone, compact, and cost-efficient multimodal microscopic system developed on a Raspberry Pi SBC platform that can be controlled and accessed from any Wi-Fi enabled device is demonstrated. Multiple imaging modalities such as BF, DF, OI, and DPC have been performed on the same system through imaging of various biological specimens. The performance of the proposed tool is found to be at par with that of a conventional laboratory microscope. The use of OLED in the present imaging system has aided in achieving multimodal imaging on a single optical setup. The developed system, when accessed remotely through an open access application on various display devices, suffers from a little delay in displaying the images due to the wireless signal lag. This can be obviated by integrating an LCD display panel directly onto the Raspberry Pi SBC module itself and developing a custom-designed application for the Raspberry Pi camera module v3. Additionally, the magnification and the resolution of the imaging system can be further improved and optimized by using the 4f optical configuration for the imaging system and by utilizing other micro-objective lenses as per the requirement. The proposed system can be seen as a cost-effective alternative imaging platform where captured images can be shared almost instantly to any part of the world through the broadband network that can be accessible from a smartphone, tablet, or PC.

Author Contributions

Biprav Chetry: conceptualization, methodology, validation, formal analysis, investigation, software, data curation, visualization, writing – original draft. **Pabitra Nath:** conceptualization, methodology, supervision, writing – review and editing, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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